- Scope for future improvements such as entropy-based scoring and common-password detection.

3.6 Core Logic

The password_validator() function is the core of the Password Strength Checker. It validates the password against five security criteria and classifies it as Weak, Medium, or Strong based on the rules satisfied. This keeps the validation logic modular, readable, and easy to extend.

```
def password_validator(password):
    has_num = any(ch.isdigit() for ch in password)
    has_lower = any(ch.islower() for ch in password)
    has_upper = any(ch.isupper() for ch in password)
    has_special = any(ch in SPECIAL_CHARS for ch in password)
    long_enough = len(password) >= 8

    if not long_enough:
        return "weak", "Password needs to have at least 8 characters.", checks
    if has_num and has_lower and has_upper and has_special:
        return "strong", "Password is Strong! Good to go.", checks   # all criteria met
    elif has_num and has_lower:
        return "medium", "Try adding an uppercase letter and a special character.",
checks
```

Note: the strength categories currently rely on fixed rule combinations rather than a weighted score; introducing an entropy- or scoring-based model is planned as a future enhancement.

2.7 Output Screenshot